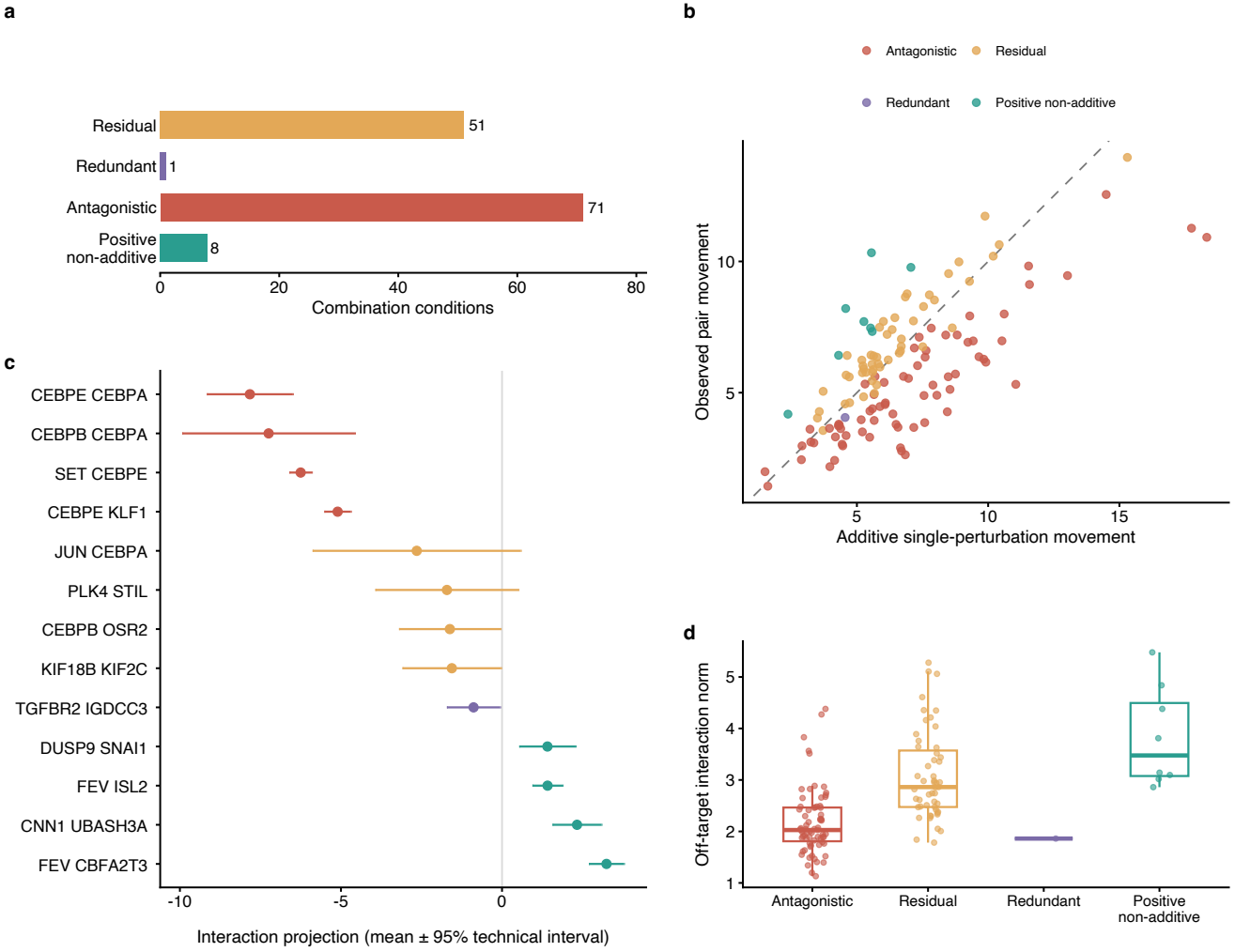
a
Residual
Redundant
Antagonistic
Positive non-additive
51
1
71
8
0
20
40
60
80
Combination conditions
b
Antagonistic
Residual
Redundant
Positive non-additive
Observed pair movement
Additive single-perturbation movement
5
10
15
c
CEBPE CEBPA
CEBPB CEBPA
SET CEBPE
CEBPE KLF1
JUN CEBPA
PLK4 STIL
CEBPB OSR2
KIF18B KIF2C
TGFBR2 IGDCC3
DUSP9 SNAI1
FEV ISL2
CNN1 UBASH3A
FEV CBFA2T3
-10
-5
0
Interaction projection (mean ± 95% technical interval)
d
Off-target interaction norm
1
2
3
4
5
Antagonistic
Residual
Redundant
Positive non-additive